(i) Outcomes:

$$\text{all } N \text{ good: } \Pr = q^N, \quad \text{gain} = N - T$$
$$\text{tested } T \text{ good, } N-T \text{ contains infected: } \Pr = q^T - q^N, \quad \text{gain} = -D$$
$$\text{a tested stick infected: } \Pr = 1 - q^T, \quad \text{gain} = 0$$

$$E_{gain} = (N-T)q^N - D(q^T - q^N) = (N - T + D)q^N - Dq^T$$
$$E_{loss} = -E_{gain} = Dq^T - (N - T + D)q^N$$

(ii)

$$\alpha = \frac{DT}{N(N-T+D)} = 1 - \frac{(N-T)(N+D)}{N(N-T+D)}$$

$$\because 0 \le T < N,\ D > 0$$
$$\therefore 0 \le \alpha < 1$$

$$\frac{dE}{dq} = N(N-T+D)q^{N-1} - DT\,q^{T-1} = q^{T-1}\Big(N(N-T+D)q^{N-T} - DT\Big)$$
$$\frac{dE}{dq} = 0 \implies q^{N-T} = \frac{DT}{N(N-T+D)} = \alpha$$
$$E = (N-T+D)q^T\alpha - Dq^T = q^T\Big((N-T+D)\alpha - D\Big)$$
$$\because q^T = \alpha^{T/(N-T)}$$
$$\therefore E_{\min} = -\frac{D(N-T)}{N}\,q^T$$

(iii)

$$\begin{aligned} E_{loss}(T+1) - E_{loss}(T) &= Dq^{T+1} - (N-T-1+D)q^N - Dq^T + (N-T+D)q^N \\ &= Dq^T(q-1) + q^N \\ &= q^N - Dpq^T \end{aligned}$$

It is increasing when $q^{N-T} > Dp$
As $p \to 0$:

$$q^{N-T} = (1-p)^{N-T} \to 1, \qquad Dp \to 0$$

$\exists\, \beta > 0$ such that for $0 < p < \beta$, $q^{N-T} > Dp$ for all T.

$$E_{loss}(T+1) - E_{loss}(T) > 0 \ \ \forall T$$

$E_{loss}(T)$ increasing, minimized at $T = 0$

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Last updated: July 25, 2026